



Research article

The influence of adjustable ring-mode laser on the formation of intermetallic compounds and mechanical properties of ultra-thin Al-Cu lap welded joints

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ABSTRACT

Aluminum-copper dissimilar welding is a highly demanded connection process; however, welding defects and the excessive growth of intermetallic compounds (IMCs) cause pose challenges for its application. This study uses an adjustable ring-mode (ARM) laser technology to achieve lap welding of ultra-thin Al-Cu plates. Lap-welding experiments were conducted using three laser modes—fixed core power, fixed ring power, and varying welding speed—to investigate the evolution of material mixing, intermetallic compound distribution, and joint strength under different modes. Our results indicate that the high energy density of the core laser is beneficial for increasing the penetration depths of joints, whereas the large action area of the ring laser is beneficial for improving the stabilities of melt pools. The joint action of the adjustable ring-mode (ARM) laser increased the melting width and depth of the joint, and the mixing of Al and Cu was controlled in the Al-Cu mixed zone at the upper part of the weld, to limit element mixing in the Cu-rich zone of the weld interface and suppress the distribution of intermetallic compounds. In addition, the ring laser induced the aluminum in the upper part of the molten pool to invade from both sides of the interface to the bottom, forming a certain Al invasion depth. This limited the accumulation of intermetallic compounds at the interface, optimized the path of shear fracture propagation, and improved the shear strength of the joint. This study provides a research basis for further exploring the material flow mechanism and optimizing the intermetallic compound distribution during the Al-Cu adjustable ring-mode (ARM) laser dissimilar welding process.

1. Introduction

Copper exhibits good conductivity, thermal conductivity, and excellent processing performance. It is widely used in electrical, automotive, and new energy fields. Aluminum, as an excellent conductor with lower density and price, has become a good substitute for copper. This is especially true in the application of tabs and busbars in new energy batteries. Therefore, the connection of copper and aluminum (i. e., dissimilar metals), especially in thin and ultra-thin plates (≤ 1 mm), is becoming increasingly common [1]. The significant differences in the physical and chemical properties of copper and aluminum cause numerous issues in joints during welding. The mutual solubility of Cu and Al is poor. Intermetallic compounds (IMCs) layers form during

solidification processes, and brittle IMCs lead to a stress concentration in the joint, rendering the joint highly crack sensitive and significantly affecting the conductivity and mechanical properties of the joint [2]. In addition, the differences in physical properties cause defects, such as spatter and porosity, to occur during the welding process [3].

Research has shown that the formation of IMCs in Al-Cu welded joints mainly involves θ -Al₂Cu, η_2 -AlCu, ζ_2 -Al₃Cu₄, and γ_1 -Al₄Cu₉ phases [4]. The presence of excessive IMCs in Al-Cu welded joints is harmful, but IMCs are the products of metallurgical bonding between the substrates. Xue et al. [5] demonstrated that a uniform and thin Al-Cu interface can enhance the strength of a joint and hinder the rapid propagation of cracks. However, in this study, the process window was narrow, and inappropriate process parameters could easily result in Cu

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pieces in the joint. There is significant stress concentration at both ends of the pieces, which can easily lead to joint fracture. Mathivanan et al. [6] found that there are two main types of intermetallic compounds (IMCs), namely Al_2Cu and Al_4Cu_9 , on the weld interface. The bonding strength of the joint was increased by controlling the thickness of the IMCs to several micrometers through process control. Zhou et al. [7] demonstrated that a thin IMC is beneficial for joint strengthening and further found that a continuous serrated layer of $\text{Al}_{4.2}\text{Cu}_{3.2}\text{Zn}_{0.7}$ is closely related to the high strength of the joint. The strength of the bonding between the IMC and base metal and that between different IMC phases determines the location and path of fracture in welded joints. Schmalen et al. [8] found that cracks appeared inside a laser-welded joint, extending from the $\text{Al}_3\text{Cu}_4/\text{Al}_4\text{Cu}_9$ interface to Al_2Cu and proved that this defect was related to the high hardness and brittleness of the various IMCs in the joint. However, the welding process was not optimized on this basis to suppress the generation of mixed IMCs in this research. Mao et al. [9] conducted in situ tensile testing on an Al-Cu joint and found that the fracture surface exhibited a typical brittle fracture. Microcracks in the joint first nucleated at the Al/ Al_2Cu interface and then continued to propagate along Al_2Cu . In addition, new microcracks formed at the Al/ Al_2Cu interface; therefore, Al_2Cu is considered one of the most dangerous IMCs in this study. This is consistent with the results of Dang et al. [10], where Al_2Cu , Al_4Cu_9 , and AlCu phases were generated in inertial friction welding joints. The fracture also propagated along the Al/ Al_2Cu interface, leading to a cleavage fracture. The scholar proves that with the increase of heat input, the Al_4Cu_9 phase is formed between Al_2Cu and Cu. The formation of Al_4Cu_9 reduces the misfit at the friction interface, which is favorable for the improvement of tensile strength of the welded joint. However, an increase in heat input can easily lead to excessive growth of lamellar IMCs, which will deteriorate the strength of the joint. In addition to directly affecting the strength of the joint, the transitional growth of the IMCs caused an increase in voids at the interface. Wang et al. [11] found that under the condition of 600 °C/9 MPa, after 10 s, significant voids began to appear near the Cu/Al brazed joint due to an increase in the Al_2Cu content, and the performance of the joint deteriorated. The content and distribution of IMCs are closely related to the strength of the joint. In particular, the Al_2Cu phase, which is easily generated during welding, strongly correlates with the crack propagation path when the joint is subjected to a load. Therefore, special attention should be paid to the Al_2Cu phase generated during aluminum-copper welding.

Owing to the bonding between the material atoms involved in the welding process, the welding of dissimilar aluminum copper materials inevitably involves mutual diffusion between copper and aluminum elements, manifesting as the unavoidable generation of IMCs in the joint. Solid-state welding techniques, such as friction stir welding, ultrasonic welding [12], and resistance spot welding [13], have been widely used to connect aluminum and copper to suppress atomic diffusion and excessive IMC generation. This is because the peak temperature of solid-state welding is lower, and the overall heat input is smaller. This inhibits the transition growth of the IMCs. In addition, owing to the low degree of metallurgical bonding, the bonding strength of the joint is low, and the welding efficiency is usually low. During fusion welding, the peak temperature is significantly higher than the melting point of the material, and the molten material forms a molten pool. Under the action of the molten pool, the material flows sufficiently. According to previous research, fusion welding of aluminum and copper includes tungsten inert gas (TIG) welding [14], vacuum electron beam (EBW) welding [15], and laser welding. Among these, laser welding is the most widely used [16]. Because laser welding has a high energy density and small heat-affected zone, achieve automation is easy. In addition, the optical components do not need physical contact with the workpiece during the welding process, making automation easier. Therefore, controlling the type and quantity of IMC in the aluminum copper melt pool during laser welding has become a research hotspot. Interface materials were added between aluminum and copper to control the metallurgical reactions

between the substrates. Yan et al. [17] used an intermediate layer of Ni to weaken the temperature diffusion at the Al-Cu interface and control the melting of Cu. The AlNi phase was found in the joint, and it refined the Al_2Cu IMC and improved the mechanical properties of the joint, proving that the Ni foil had a certain inhibitory effect on the generation of the IMC in aluminum copper laser welding. However, the Ni foil did not eliminate the existing defects such as cracks and pores in the weld seam. Additionally, increasingly advanced laser control processes have been introduced to improve the joint performance [18]. The applications of laser composite welding [19] and electromagnetic field-assisted laser welding [20] have increased gradually. Ma et al. [21] combined preheating (PH), active welding (AW), and cooling (CL) waveforms using laser shaping to create six-pulse shapes for Al-Cu welding. The results showed that pulse shapes of the PH and AW phases had the best joint performance; however, the welding speed used in the experiment was relatively low, and the welding efficiency was not high. In addition, obvious cracks and pores were present in these joints, and the cracks were related to Al_2Cu and Al_4Cu_9 components formed at the boundary of the melt pool. Thus, adding intermediate layers and adjusting the process parameters play certain optimization roles in aluminum copper laser welding. However, traditional lasers can reduce the heat distribution only through overall power adjustment and waveform control. In recent years, the use of adjustable ring mode (ARM) lasers has increased gradually. This mode can be used to independently adjust the laser power of the core and ring spots. This can control the heat input and distribution more accurately, thereby reducing the temperature gradient of the molten pool and improving the flow of the molten pool [22]. Currently, research on point-ring lasers in dissimilar aluminum and copper welding is relatively scattered. Cha et al. [23] used an ARM laser to weld Al-Cu joints and explored the surface formation of welds by adjusting the power ratio of the core and ring. The results showed that increasing ring power increased the surface width of the weld and reduced the number of pores in the weld. However, current research on the mixing behavior of aluminum and copper under the combined action of ARM lasers is relatively limited, and the influence of materials and IMC distribution at the interface under different flow conditions remains unclear.

This research aims to investigate the effect of the ARM laser on lap joints of 0.5 mm thick Al and Cu plates and explore the flow of Al and Cu under different process combinations. Furthermore, we analyze their influence on the IMC distribution and joint strength using three trial groups (by changing ring power, point power, and welding speed). The evolution of weld geometry, material mixing, mechanical properties, and electrical conductivity were evaluated, revealing the upwelling of copper guided by the recoil pressure and downward invasion process of aluminum under the Marangoni force of the molten pool under the ARM laser. We suppressed the types of IMCs at the interface and improved the strength of the joint by controlling the power of the core and ring lasers, which provided research ideas for the further application of ARM in Al-Cu welding.

2. Experimental procedure

2.1. Materials and methods

In this study, AA1060 pure aluminum (purity over 99.6 wt%) and T2 pure copper (purity over 99.8 wt%) ultra-thin plates, both with dimensions of $100 \times 50 \times 0.5$ mm, were used. Prior to welding, the surfaces were treated with acetone and alcohol. The welding experiment was conducted using an ARM fiber laser with a maximum total power (P_T) of 6 kW, by following an Al-Cu (aluminum on the upper side and copper on the lower side) lap welding method. The welding assembly diagram is shown in Fig. 1a. The diameters of the center and ring beams of the ARM fiber laser were 50 and 150 μm , and they had maximum powers of 4 and 2 kW, respectively, and the schematic diagram of the spot size is shown in Fig. 1b. Here, the power ratio r is defined as the

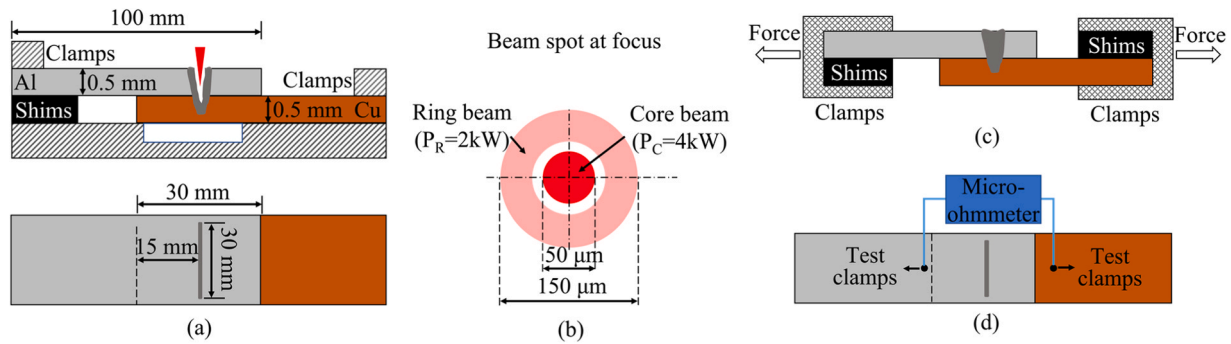


Fig. 1. The schematic diagrams of experiment process. (a) Welding and fixture device. (b) Adjustable ring-mode (ARM) laser spot. (c) Shear testing. (d) Resistance testing.

ratio of the core power (P_C) to the ring power (P_R). This means that when r is ∞ , it represents only the center beam. The entire welding experiment did not use a filler wire or protective gas, and the incidence angle of the laser beams was tilted by 5° . The welding process adopted zero defocus, which means that the distance between the focus of the laser speckle and surface of the Al plate was 0 mm.

To study the microstructure and performance evolution of the Al-Cu lap joints under different power ratios and welding speeds, three trial groups were designed by separately varying the ring power (P_C fixed at 1000 W), core power (P_R fixed at 1500 W), and welding speed (P_C and P_R fixed at 1000 and 2000 W, respectively). The welding parameters in each group, such as the laser power, power ratio, and welding speed, are listed in Table 1. Each welding experiment was repeated five times.

2.2. Microstructure characterization

After welding, the weld formation and surface spatter were observed using a stereo microscope and confocal scanning laser microscope (CSLM; KEYENCE VK-X3000, China). The CSLM was equipped with objective lenses of different magnifications. Considering the high roughness of the weld surface, $5\times$ objective lens with a field of view (FOV) of $2770 \times 2076 \mu\text{m}^2$ was used. Subsequently, two weld seam segments were selected at the middle position of each weld seam for height imaging, and the average surface roughness (S_a) was calculated.

Table 1

Laser welding parameters. The naming method of trial ID is based on the core power, ring power, and welding speed.

Trial group	Trial ID	Core power (W), P_C	Ring power (W), P_R	Power ratio, $r = P_C/P_R$	Welding speed, v (mm/s)
Change ring power	C-R0	1000	0	∞	250
	C-R250	1000	250	4	250
	C-R500	1000	500	2	250
	C-R750	1000	750	1.33	250
	C-1000	1000	1000	1	250
	R1000	1000	2000	0.5	250
	C-R2000				
Change core power	C800-R	800	1500	0.53	250
	C920-R	920	1500	0.61	250
	C1040-R	1040	1500	0.69	250
	R-1160	1160	1500	0.77	250
	C1160-R	1280	1500	0.85	250
	C1280-R				
	R-1280				
Change welding speed	CR-200	1000	2000	0.5	200
	v200	1000	2000	0.5	300
	CR-300	1000	2000	0.5	350
	v300				
	CR-350				

The metallographic samples were cut in the middle of the weld along the vertical welding direction and encapsulated in acrylic resin. After demolding, the samples were completely coarse and fine ground using 600 #, 800 #, 1000 #, 1500 #, 2000 #, and 3000 # sandpapers. Subsequently, they were polished using a $0.5 \mu\text{m}$ diamond suspension. The phases were determined using X-ray diffraction (XRD) (D8 ADVANCE Da Vinci, Germany) and an electron backscatter diffraction (EBSD) instrument equipped with a scanning electron microscope (MIRA3, Czech Republic). XRD was performed at an accelerating voltage, a current, and a scanning rate of 40 kV, 40 mA, and $3^\circ/\text{min}$, respectively, within a testing range of 20° – 100° . The specimens used for EBSD testing were first polished to a mirror-like finish and then subjected to 2 h of vibration polishing. AZtecCrystal 2.1 software was used to process the EBSD data. The microstructures and morphological characteristics of the precipitates were observed using scanning electron microscopy (SEM; Tescan VEGA3, Czech Republic) equipped with backscattered electron (BSE) and energy dispersive X-ray spectroscopy (EDS). The size characteristics of the welded joints were measured using an image analysis software. Finally, the microstructure of the weld cross-section, including the mixing of Al and Cu elements, Al-Cu interface characteristics, and distribution of IMCs in the joint, was characterized.

2.3. Mechanical testing

To compare the shear strength of the joints under different welding parameters, the samples were tested using a universal tensile testing machine (Zwick Z020, Germany) in displacement control mode at a tensile rate of 1 mm/min. A tensile diagram is shown in Fig. 1c. For each welding parameter, the shear strength value was the average shear strength of three specimens. After the shear test, the morphologies of the fractures on both sides of aluminum and copper were observed by SEM to determine the distribution of the IMCs on the fracture surface and reveal the fracture type and path.

2.4. Resistance testing

To investigate the difference in the conductivity of the joints, the resistance of the lap joints was measured using a micro-ohmmeter (GWINSTEK GOM-805, China) at room temperature. The ohmmeter was based on the four-wire Kelvin principle, and it had an accuracy of $0.1 \mu\Omega$. A test clamp was fixed on the substrate outside the aluminum-copper overlap. The clamping diagram of the testing process is shown in Fig. 1d. To improve the testing accuracy, the resistance values of each parameter were recorded after 20 s of testing, once the reading had stabilized, and the average test value of three samples was taken.

3. Results

3.1. Weld morphology characteristics

The weld formation and surface height diagrams are shown in Fig. 2a. Fig. 2b shows the roughness (S_a) of the area marked with white dashed lines. According to the statistical results, as the power ratio r decreased, S_a also decreased. Similar to S_a , as r decreased, the spatter on the weld surface also decreased, as shown in Fig. 2c. The ARM laser can reduce the temperature gradient of the molten pool and thereby improve the stability of the welding process and surface formation of the weld seam. When r was 0.5, red scattered data indicated that the surface roughness first decreased and then increased with an increase in welding speed. According to Wan et al. [24], changes in welding speed can cause Rayleigh instability, leading to fluctuations in the melt pool and formation of irregular joint shapes.

Fig. 3 shows a BSE image of the weld cross-section, with white dashed lines indicating the weld boundary. The surface weld width (w_{top}), interface weld width (w_{int}), and penetration depth (d) of the weld cross-section were statistically analyzed, and the results were obtained by taking the average of the parameters of the two samples, as shown in Fig. 4. No obvious pores or large cracks were observed in any of the joints. All the welds exhibited a funnel-like structure with a wider top and narrower bottom, indicating that the melting of aluminum was significantly higher than that of copper. The melting amount of aluminum significantly increased with an increase in the core and ring powers, whereas the influence of the ring power on the melting of copper was relatively small. According to Fig. 4, when the core power is below 1000 W or welding speed is above 250 mm/s, the penetration is less than 70 μm . As the core power increased, the amount of melted

copper increased, and the interface weld width increased significantly. Thus, the melting amount of copper primarily depends on the core power because the energy density of the core power is more concentrated with stronger penetration, and it is more likely to cause deep melting in the weld seam [25], leading to increased penetration depth and melting of copper. Under the joint action of the ARM laser, the upper aluminum plate melted first, and the core laser continuously increased the weld penetration depth. The ring power expanded the width of the molten pool, increased the high-temperature area, reduced the temperature gradient, and alleviated the instability of the molten pool caused by the temperature changes and molten metal flow, thereby improving the stability of the molten pool [26].

3.2. Element mixing and intermetallic compounds (IMCs) distribution characteristics

The mixture of Al and Cu and distribution of IMCs in the joint were characterized using SEM-EDS. Fig. 5 shows the elemental surface scanning results for some of the joints, where red and green colors indicate Al and Cu, respectively. The mixing of Al and Cu occurred in three regions of the weld cross-section, including the upper Al side, middle Al-Cu interface, and lower Cu side. Among these the mixing of the Al-Cu interface was the most intense. The degree of mixing primarily depended on the melting amount of the base metal and flow of the molten pool, which in turn determined the IMC distribution at the Al-Cu interface. Therefore, when the penetration depth is significantly small, copper melts insufficiently and element mixing does not occur, as in C820-R. For joints with large penetrations, such as C-R0, the melting amount of Cu increased, and the mixing area of Al and Cu increased.

Mixtures of aluminum and copper eventually generate various IMCs

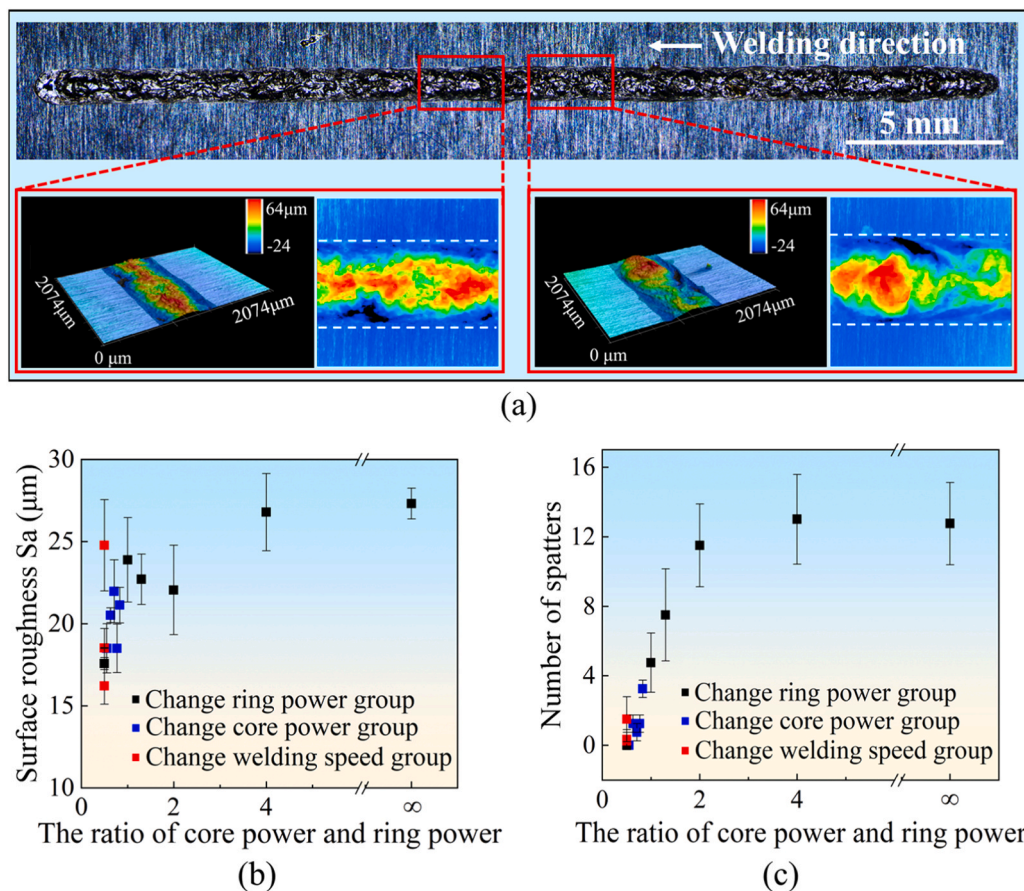


Fig. 2. Weld and surface quality. (a) Weld morphology. (b) Statistical results of weld roughness. (c) Statistical results of weld spatters. The error bar represents the standard deviation of the mean value.

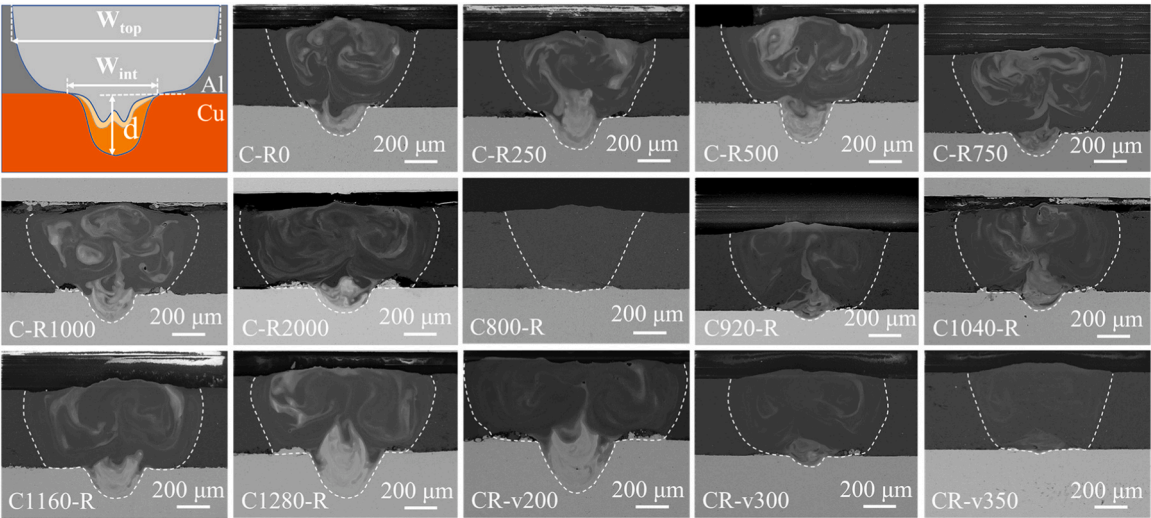


Fig. 3. Cross section backscattered electron (BSE) images of weld seams. The schematic diagram in the upper left corner is used to represent the meaning of weld size.

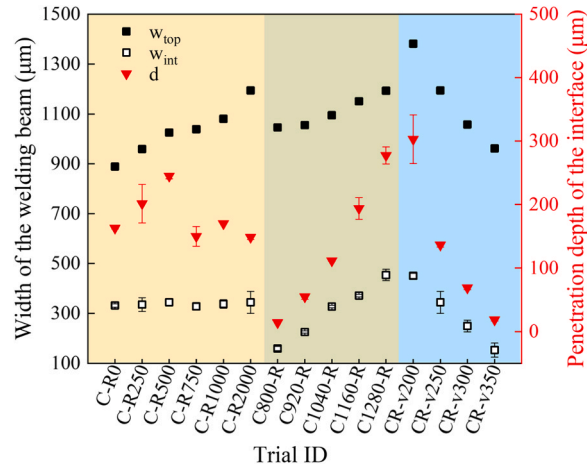


Fig. 4. The statistical results of weld size. The error bar represents the standard deviation of the mean value.

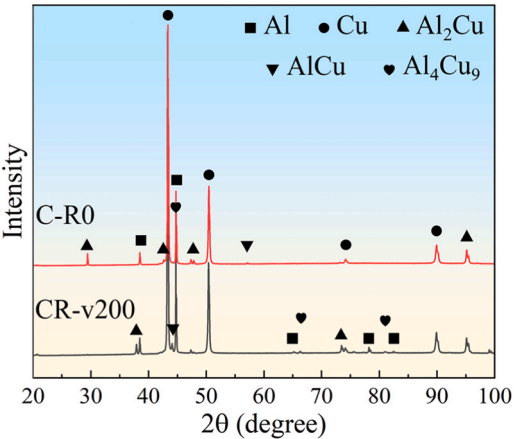


Fig. 6. X-ray diffraction (XRD) profiles. The red spectral line represents C-R0 sample, and the black line represents CR-v200 sample.

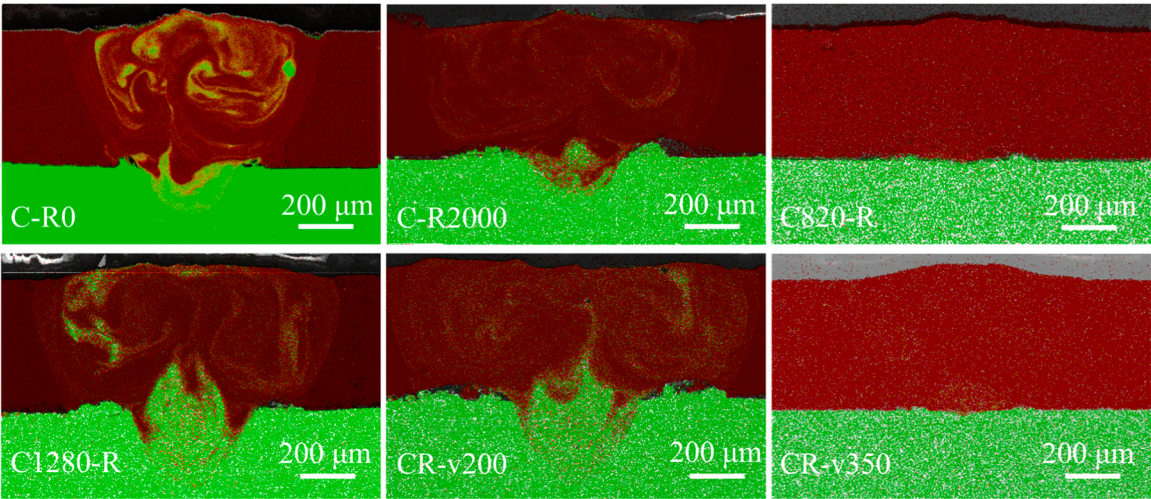


Fig. 5. The Al and Cu elements map scanning results of weld cross section. This figure selects two joints with larger and smaller penetration depth in three groups of welding trials.

in the weld seam. Fig. 6 shows the diffraction peaks obtained by the XRD characterization of the C-R0 and CR-v200 welding cross-sections. In addition to the diffraction peaks of the base materials, Al_2Cu , AlCu , and Al_4Cu_9 phases were detected in the joint, with Al_2Cu exhibiting the highest diffraction peak, indicating that the easy generation of this phase in the joint. Compared with C-R0, the CR-v200 sample has more AlCu and Al_4Cu_9 phases.

Owing to the melting of Al in all the samples, the melting amount of Cu in the joint was closely related to the element mixing and IMC distribution. To identify the differences between the trial groups, one joint was selected from each group for IMC characterization. A large penetration depth promotes the mixing of elements, which is conducive for revealing the IMC distribution characteristics. Therefore, the sample with the largest amount of melted Cu in each group was selected. In addition, to facilitate a comparison with core-ring coupling, the pure core power-welded joint C-R0 was selected from the first group, and the samples with the largest penetration depths, namely C1280-R and CR-v200, were selected from the other two groups. These three samples satisfied the law of increasing amount of copper melting. Because the Al-Cu interface, which is the main area connecting the matrix, has a decisive influence on the performance of the joint, the IMCs in this area were characterized.

The Al-Cu interface morphology and IMC distribution in the C-R0 sample are shown in Fig. 7 and Table 2, respectively. The line scan results along the penetration direction are shown in Fig. 7b. The results indicate the presence of two IMC regions rich in Al and Cu at the interface. Fig. 7c,e show that the morphologies on both sides of the interface are symmetrical; that is, the IMCs have a similar distribution, with Cu-rich IMCs, including the white cluster AlCu phase and Al_4Cu_9 phase, being generated close to copper. The Al_2Cu phase primarily formed near aluminum and extended to the Al-Cu eutectic zone in the form of dendrites. Owing to the interaction between aluminum and copper at the weld root, an IMC mixed zone with Al_2Cu , AlCu , and Al_4Cu_9 was formed, as shown in Fig. 7d. EBSD was used to scan the

phase distribution of the white-dashed zone (Fig. 7a), and the results are shown in Fig. 8a. These results were consistent with the XRD results shown in Fig. 6. The IMCs contained Al_2Cu , AlCu , and Al_4Cu_9 , with the Al_2Cu phase being the most abundant and forming a flaky phase near the aluminum side. Because of the low heat input and melting depth, the content of Al_4Cu_9 phase was relatively low.

Fig. 9 and Table 3 present the characterization results of the Al-Cu interface of the C1280-R sample. Owing to the increase in the point and ring powers, the penetration depth and interface width increased significantly. An obvious Cu-rich zone was observed, mainly consisting of an Al_4Cu_9 phase, at the bottom of the weld, and it was distributed as shown in Fig. 9d. The results in Fig. 9c,e show that the flow of aluminum to the weld root is limited and primarily concentrated on both sides of the interface. Thus, obvious aluminum distribution areas form in these two places, which in turn lead to the formation of an Al-Cu eutectic, Al_2Cu , and a small amount of AlCu phases. The dendrites of the Al_2Cu phase showed more slender characteristics, with an obvious needle-like transition between the Al-Cu eutectic and Al.

The interface morphology of the CR-v200 sample is shown in Fig. 10. This sample had the largest heat input; thus, its penetration depth and interface width were also the largest. Fig. 10c,e showed that the content of Al_2Cu in the sample increased, and a small amount of Al_2Cu sporadically formed in the Al-Cu eutectic. The dendrite characteristics of the Al_2Cu phase were more evident, and the length and width of the dendrites increased. Simultaneously, AlCu and Al_4Cu_9 phases were also detected. This was because the welding speed was the slowest, and the cooling time of the molten pool increased, providing sufficient time for the diffusion of elements and growth of the IMCs. A crack propagating along the IMC clusters at the Al-Cu interface is shown in Fig. 10e. The increased heat accumulation during the welding of the sample and large difference in the thermal expansion coefficients of the mixed IMCs resulting in a large welding residual stress [27]. The cracking finally stopped in the island-shaped Al_4Cu_9 phase, indicating that the mixed IMCs on both sides of the interface were sensitive to cracking and that

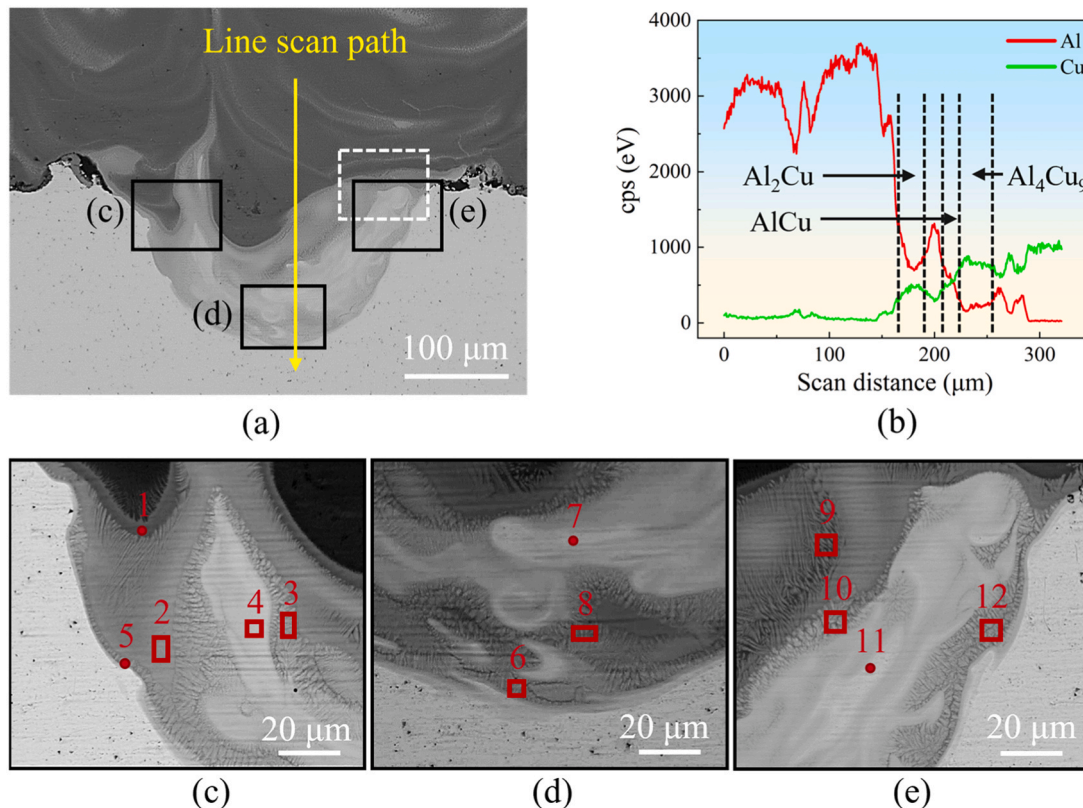


Fig. 7. Microstructure of C-R0 sample. (a) Al-Cu interface. (b) Energy dispersive X-ray spectroscopy (EDS) element line scanning. (c-e) Local magnification.

Table 2
The elements composition of intermetallic compounds (at%) in Fig. 7.

Points	1	2	3	4	5	6	7	8	9	10	11	12
Al	81.62	62.61	59.85	37.55	51.28	48.82	23.76	57.94	74.48	53.31	30.24	57.61
Cu	18.38	37.39	40.15	62.45	48.72	51.18	76.24	42.06	25.52	46.69	69.76	42.39
Phase	Al-Cu eutectic	Al ₂ Cu	Al ₂ Cu + AlCu	Al ₄ Cu ₉	AlCu	AlCu	Cu+ Al ₄ Cu ₉	Al ₂ Cu + AlCu	Al+ Al ₂ Cu	Al ₂ Cu + AlCu	Al ₄ Cu ₉	Al ₂ Cu + AlCu

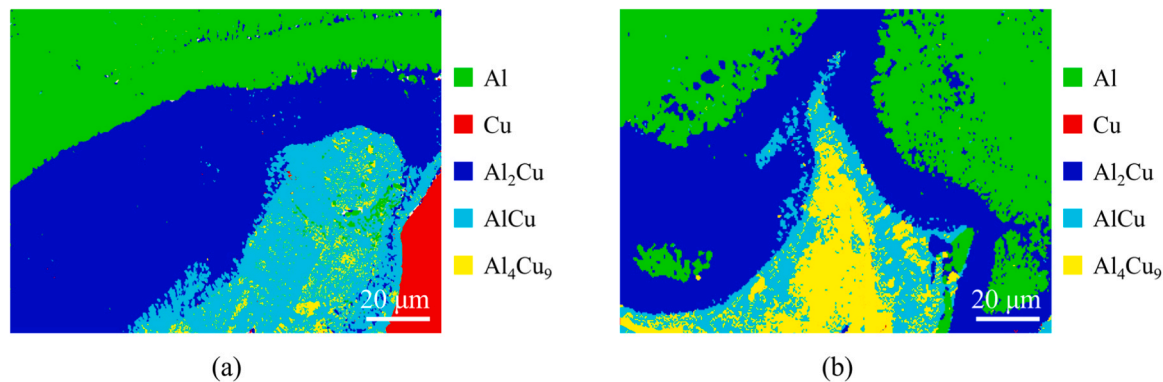


Fig. 8. The phase distribution results of electron backscatter diffraction (EBSD). (a) C-R0 sample. (b) CR-v200 sample.

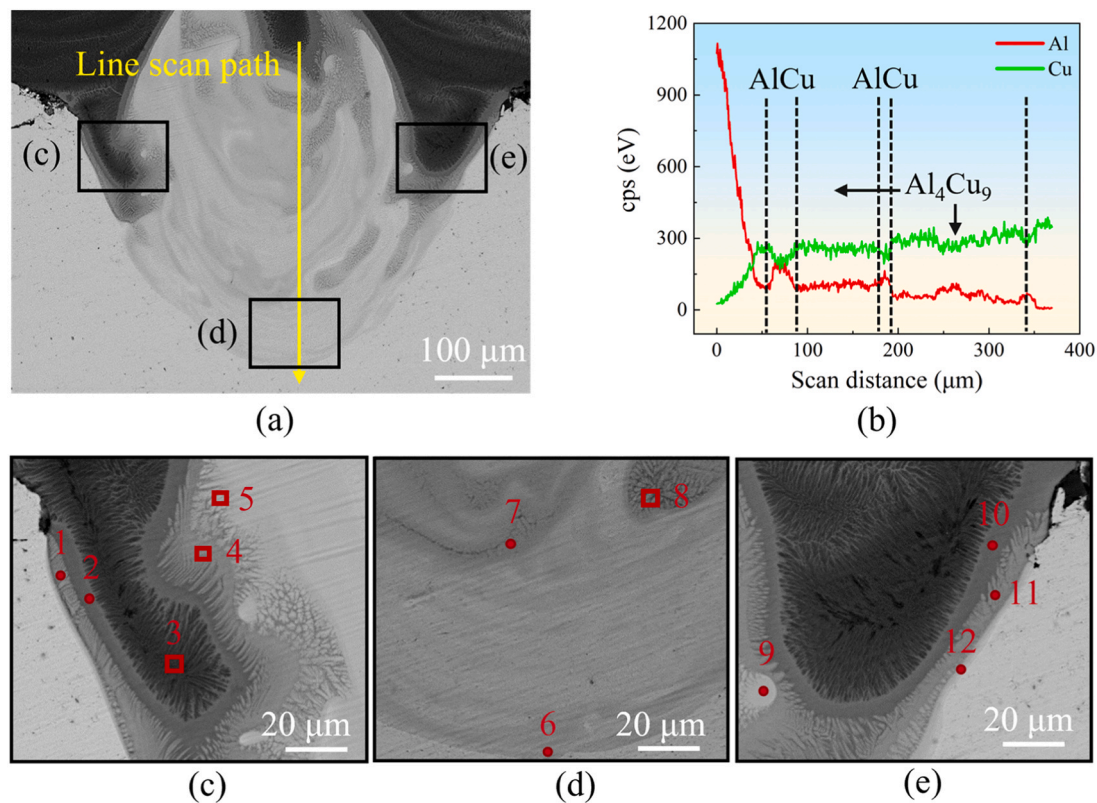


Fig. 9. Microstructure of C1280-R sample. (a) Al-Cu interface. (b) Energy dispersive X-ray spectroscopy (EDS) element line scanning. (c-e) Local magnification.

Table 3
The elements composition of intermetallic compounds (at%) in Fig. 9.

Points	1	2	3	4	5	6	7	8	9	10	11	12
Al	69.84	83.73	95.61	70.13	56.02	34.64	45.16	56.48	36.15	83.96	70.89	25.97
Cu	30.16	16.27	4.39	29.87	43.98	65.36	54.84	43.52	63.85	16.04	29.11	74.03
Phase	Al ₂ Cu	Al-Cu eutectic	Al	Al ₂ Cu	Al ₂ Cu + AlCu	Al ₄ Cu ₉	AlCu	Al ₂ Cu + AlCu	Al ₄ Cu ₉	Al-Cu eutectic	Al ₂ Cu	Cu+ Al ₄ Cu ₉

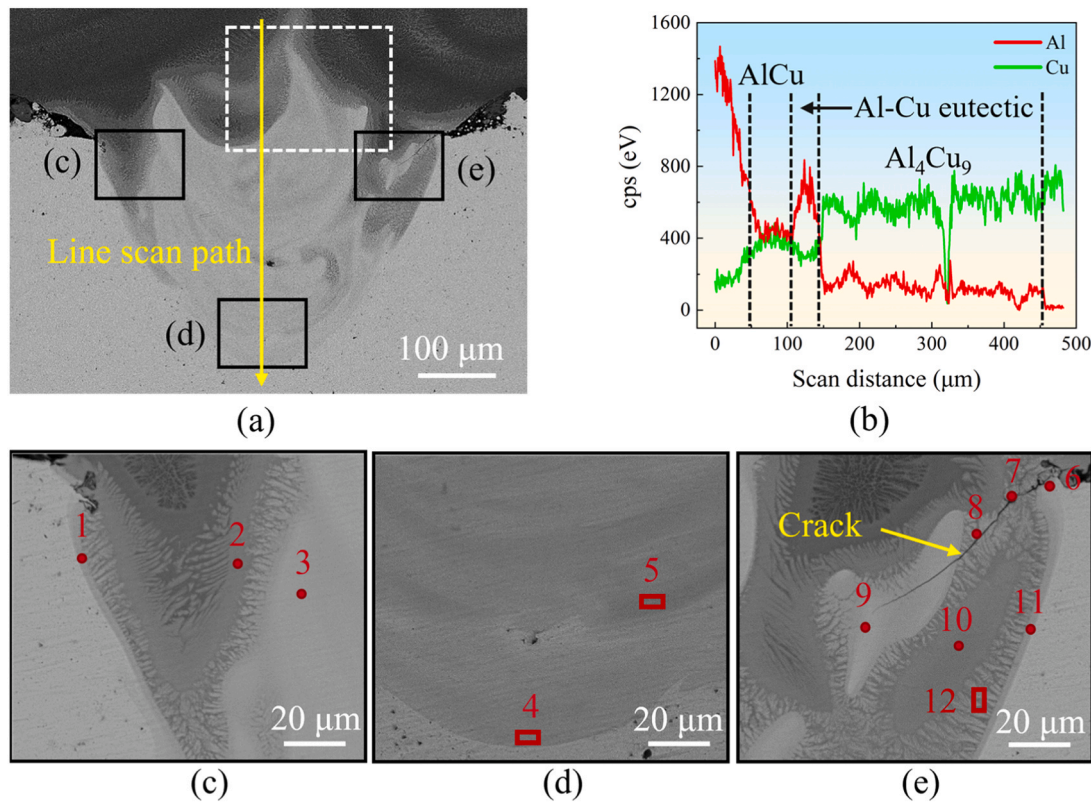


Fig. 10. Microstructure of CR-v200 sample. (a) Al-Cu interface. (b) Energy dispersive X-ray spectroscopy (EDS) element line scanning. (c-e) Local magnification.

the island-shaped Al_4Cu_9 phase suppressed cracking to some extent. The EBSD result for the white-dashed zone in Fig. 10a is shown in Fig. 8b. Compared to the C-R0 sample, this result shows more Al_4Cu_9 owing to its higher heat input and lower welding speed. In addition, the distribution of the AlCu phase was between those of the Al_4Cu_9 and Al_2Cu phases.

Based on the above analysis, when the penetration is significantly small, such as in C820-R, the amount of copper melting is insufficient, and the mixing degree of Al and Cu elements in the joint is significantly light. Furthermore, almost no IMC forms in the Al-Cu interface zone. When the melting amount of copper increases, such as in C-R0, elemental mixing intensifies, and Al_2Cu , AlCu, and Al_4Cu_9 phases form at the interface. With a further increase in the amount of melted Cu, such as in C1280-R, an Al-rich zone surrounded by Al_2Cu and a Cu-rich zone concentrated in Al_4Cu_9 form at the Al-Cu interface.

3.3. Shear strength and conductivity

Fig. 11a shows the shear strengths of the joints with different trends. In the first group, owing to the low core power, the penetration ability of the ARM laser was weak. Although the ring power gradually increased, the melting depth did not change significantly, and the effect of the ring power was not reflected. As the ring power increased, the shear strength decreased from 807 N to 505 N. This process is discussed further in the following section. In the second group, as the point power gradually increased from 800 W to 1280 W, the shear force gradually increased

from 508 N to 1030 N. The increase in the joint strength was related to the increase in the interface melting width and penetration depth. When the welding speed was varied, the sample with the lowest welding speed had the largest interface melting width and depth, and the shear force reached 1080 N. On increasing the welding speed, the strength rapidly decreased to 508 N, while the shear strength gradually increased to 652 N.

The resistance of the joint is shown in Fig. 11b. The resistance values of the first group of experiments changed little, whereas those of the other two groups fluctuated significantly. For the Al-Cu lap joints, although the non-welded zone had physical contact, the resistance value was much larger than that of the metallurgically bonded welded zone. Therefore, the current at the joint mainly flowed along the welded zone. The relationship between the Al and Cu interface connection width and resistance value is shown in Fig. 11c. Except for the points circled by red arrows, the other samples showed strong regularity; that is, with an increase in the interface width, the resistance value gradually decreased, and the decreasing trend slowed gradually. This is because when the interface width is significantly small, forming an effective anchored joint at the Al-Cu interface is challenging, and the conductive path is narrow. When the interface melting width increased, the current path became wider, and the resistance decreased. However, when there were obvious cracks at the interface, the current flow was hindered, and the resistance increased.

Table 4

The elements composition of intermetallic compounds (at%) in Fig. 10.

Points	1	2	3	4	5	6	7	8	9	10	11	12
Al	45.05	65.68	29.91	27.28	39.29	39.80	63.89	51.40	30.12	65.65	37.50	55.28
Cu	54.95	34.32	70.09	72.72	60.71	60.20	36.11	48.60	69.88	34.35	62.50	44.72
Phase	AlCu	Al_2Cu	Cu+	Cu+	Al_4Cu_9	Al_4Cu_9	Al_2Cu	AlCu	Al_4Cu_9	Al_2Cu	Al_4Cu_9	$\text{Al}_2\text{Cu} + \text{AlCu}$

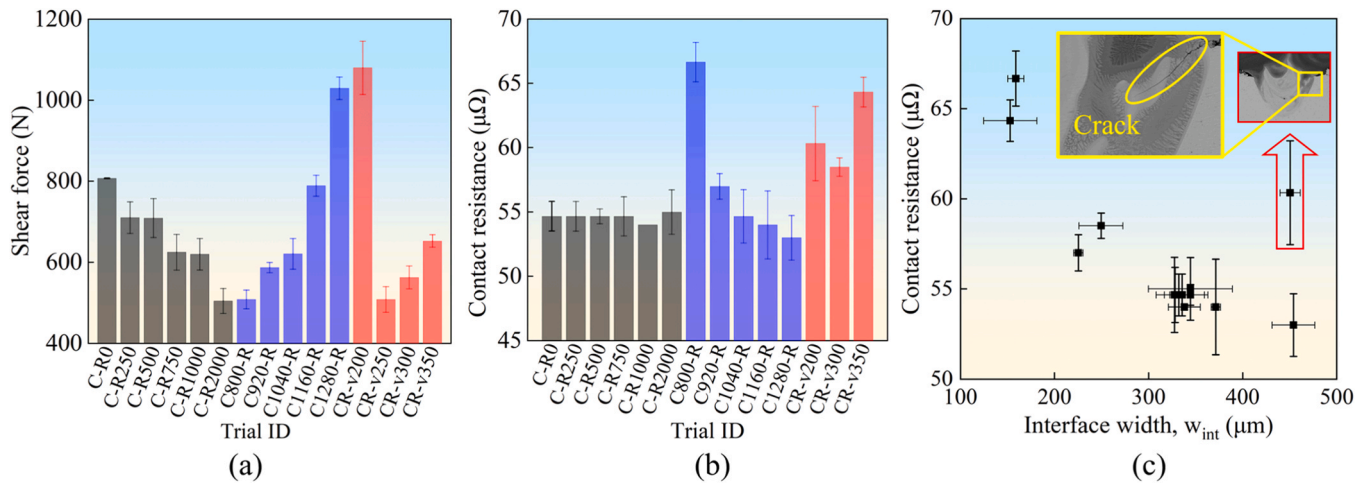


Fig. 11. Properties of welded joints. (a) Shear strength. (b) Resistance. (c) Relationship between resistance and interface width. The error bar represents the standard deviation of the mean value.

4. Discussion

4.1. Molten pool flow and element mixing behaviors

During the laser welding process, Al and Cu were melted and fully mixed under the action of the driving force of the molten pool. The driving force of the molten metal includes the vapor recoil pressure, surface tension, and Marangoni stress caused by the surface tension gradient (including the temperature gradient capillary stress and element concentration gradient capillary stress) [27]. These forces are affected by the laser energy, molten pool temperature, and solute concentration; therefore, the dominant forces acting on liquid metals vary in different regions or moments. Fig. 12 shows a schematic of the molten pool flow under different laser energy combinations, which was used to reveal the force of the molten pool and flow of Al and Cu. Under the action of the laser, a molten pool was quickly established, and the bottom was subjected to a high steam recoil pressure, which was perpendicular to the molten pool boundary and pointed to the outside (denoted by a blue arrow). The pressure pushed the liquid copper at the bottom to flow downward, which was hindered by the bottom and boundary of the molten pool. Hence, its trajectory changed, driving the molten Cu to form an upward eddy current (orange arrow). At this point, owing to the influence of copper upwelling, the aluminum in the melt displaced towards the top of the melt. The upper part of the molten pool was affected by the Marangoni stress (indicated by a yellow arrow) owing to the difference in the temperature gradient and solute concentration gradient. The liquid copper flowing from the bottom was rapidly pushed to the edge of the molten pool at lower concentrations and temperatures. During this process, molten aluminum was carried, and it flowed

towards both sides of the molten pool (gray arrows). Owing to the obstruction of the molten pool boundary, further flow of Cu was limited, and the unsolidified Cu flow reversed to the central area of the molten pool. The molten aluminum was also influenced, causing it to flow towards the center and bottom of the melt pool. With the departure of the laser, the molten pool rapidly cooled and solidified, yielding a weld seam with an alternating distribution of aluminum and copper vortex characteristics. Finally, there were Al-Cu mixed and Cu-rich zones at the top and bottom of the weld seam, respectively.

Owing to the differences in the laser energy and welding speed, there were some differences in element mixing. When there is only a core power, such as for the C-R0 joint in Fig. 3, the penetration depth and size of the Cu-rich zone are small, and Al easily diffuses into the weld root, making the Cu-rich zone contain more Al and resulting in the violent mixing of elements in the zone. A mixed IMC is generated as the melt pool cools. In this joint, because of the large temperature gradient and rapid cooling speed in the upper part of the molten pool, the Marangoni stress acts only for a short duration; therefore, the flow of copper in the Al-Cu mixed zone is insufficient, as shown in Fig. 12a. If a ring laser with a shallow penetration depth is introduced into this type of weld, the excess heat causes more intense element mixing throughout the joint. Ultimately, a phenomenon of multiple IMCs mixing will occur at the interface and deteriorate the strength of the joint. This corresponds to the BSE morphology (Fig. 3) and shear strength (Fig. 11a) of the first trial group. When the core power increases, and the ring laser is introduced, the melting amounts of copper and aluminum increase significantly, as in the C1280-R sample in Fig. 3. Under a stronger recoil pressure, copper accumulates in the center of the molten pool, rendering the diffusion of aluminum into the interior challenging. At this point, a

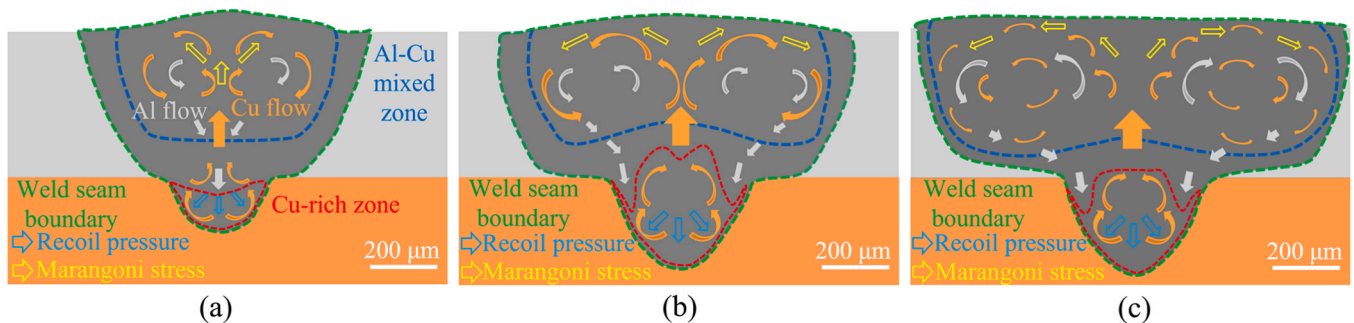


Fig. 12. The characteristic of molten pool and metal flow diagrams. (a) Only core laser joint. (b) The adjustable ring-mode (ARM) joint. (c) The adjustable ring-mode (ARM) joint with low welding speed.

larger Cu-rich zone forms at the bottom. Owing to the influence of the ring laser surrounding the central core laser, the high-temperature area of the melt pool increases, and the temperature gradient of the melt pool decreases, resulting in a long cooling time. In this case, the upwelling Cu experiences a more persistent Marangoni stress in the upper part of the molten pool, causing it to flow further and for a longer period, increasing the size of the Al-Cu mixed zone. Under the action of the copper flow, the aluminum flows (gray arrows) from both sides towards the bottom of the pool are affected by the ring laser. As the flow time increases, Al forms a significant area of invasion near the Cu-rich zone, as shown in Fig. 12b. When the welding speed is further reduced, a joint with deeper penetration is obtained, as in the CR-v200 sample. The molten-pool flow and elemental mixing processes are shown in Fig. 12c. At this time, the cooling time of the molten pool further increases, and aluminum and copper undergo more thorough mixing and flow in the upper part; thus, the size of the Al-Cu mixed zone further increases, almost coinciding with the weld boundary. Thus, under the same influence as the ring laser, more aluminum flowed from the side to the bottom of the pool, forming an invasion zone.

4.2. Types and characteristics of intermetallic compounds (IMCs) in joints

In essence, intermetallic compounds are formed by a diffusion reaction between these two metals. According to the Al/Cu binary equilibrium phase diagram, metallurgical reactions can generate approximately five types of Al-Cu IMCs phases: Al_2Cu , AlCu, Al_3Cu_4 , Al_2Cu_3 , and Al_4Cu_9 . Because Al/Cu dissimilar welding is a nonequilibrium process, only a few of these IMCs can be formed, and the formation sequence can be predicted using the Gibbs free energy criterion in the nonequilibrium state [28]. Mao et al. [9] identified that under welding conditions, the Al_2Cu and Al_4Cu_9 phases in Al/Cu joints have the most negative effective Gibbs free energy of formation (ΔG_{eff}) on the Al and Cu sides, respectively. Often, the phase corresponding to the most negative ΔG_{eff} will be preferentially generated in the joint. Haidara et al.

[29] deposited Fe, Cu, and Al on silicon oxide crystal elements and used in-situ X-ray diffraction to identify the phase at room temperature to 400 °C and 150–600 °C. Under these two temperature conditions, Al_2Cu and Al_4Cu_9 phases formed first, and then the AlCu phase was formed by the reaction of Al_2Cu and Al_4Cu_9 . Combined with the previous interface characterization results, the distribution of the IMC in the joint has a strong relationship with the Cu-rich zone during the element mixing process. The Cu-rich region preferentially generated Al_4Cu_9 , and the copper content gradually decreased outward. Furthermore, the Al-rich IMC appeared until it was replaced by Al_2Cu and Al-Cu eutectics. A schematic of the IMC formation in the micro area of the Al-Cu interface is shown in Fig. 13. Fig. 13a shows that under the action of the laser, the Al and Cu substrates changed from the solid phase (S) to the liquid phase (L) and diffused into each other. Fig. 13b shows that the Al_2Cu and Al_4Cu_9 phases were first generated on the Al and Cu sides via the interactions between the Al and Cu atoms. With the diffusion between the Al and Cu atoms, the IMCs continued to grow, as shown in Fig. 13c. Finally, the AlCu phase was generated through a reaction between the Al_2Cu and Al_4Cu_9 phases. Among them, Al_2Cu exhibits dendritic connections with AlCu and Al-Cu eutectic structures; AlCu appears as white clusters or columns, whereas Al_4Cu_9 has no specific form.

4.3. Relationship between shear strength and interface characteristics

As a connection area, the Al-Cu interface determines the shear strength, and a large number of IMCs are present at the interface. Studies have shown that there is a relationship between the interfacial bonding strength and lattice mismatch between the IMC/matrix and different IMCs [30]. According to the results demonstrated by Peng et al. [31], a small mismatch is beneficial for the bonding strength of the interface, whereas a high lattice mismatch increases the strain-free energy and reduces the bonding strength. He et al. [32] calculated the shear moduli of Al_2Cu , AlCu, and Al_4Cu_9 in aluminum-copper compounds using first-principles calculations to be 47, 67, and 70 GPa, respectively. Typically, smaller material shear moduli indicate poor shear resistance.

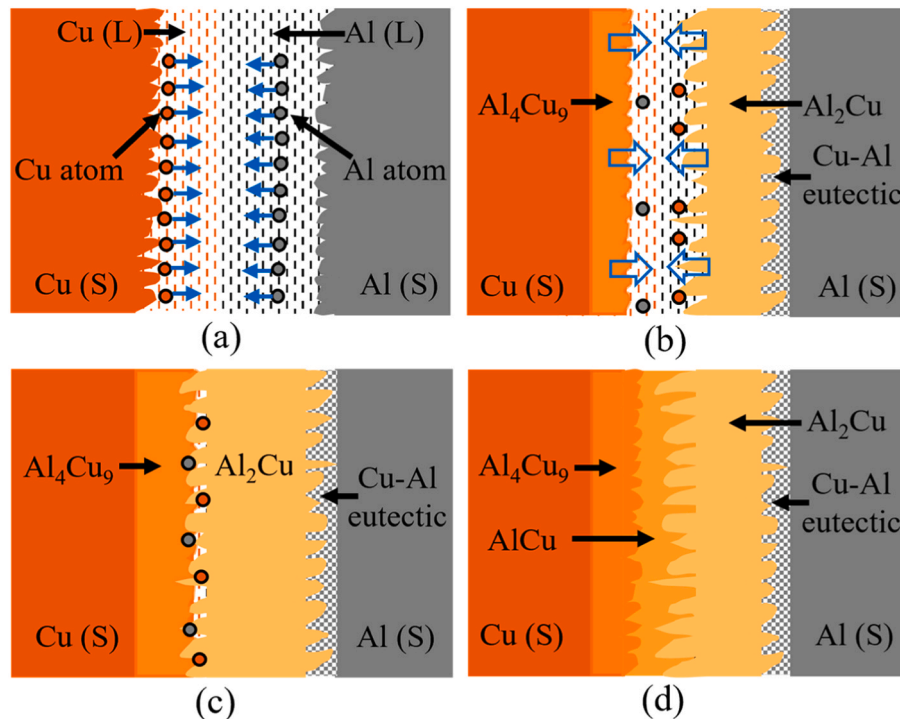


Fig. 13. The schematic diagram of intermetallic compounds (IMCs) formation in Al-Cu joint. (a) The matrix melts and diffuses into each other. (b) Al_2Cu and Al_4Cu_9 form. (c) Al_2Cu and Al_4Cu_9 continue to grow. (d) AlCu forms. The blue arrow represents the direction of atomic motion and compound growth, while the black arrow is only used for indicating purposes.

Yu et al. [33] calculated the lattice mismatch between Al/Al₂Cu, Al₂Cu/Al₄Cu₉, and Cu/Al₄Cu₉ interfaces using the lattice mismatch formula between crystals under a transmission electron microscope. The mismatches between the interfaces were 0.567, 0.586, and 0.268, respectively. Owing to the difference in the thermal expansion coefficients of the base metal and IMC, there is often a large stress concentration in the interface area of the dissimilar joint. Studies have shown that the difference in thermal expansion coefficients between the Al matrix and Al₂Cu is greater than that between Al₄Cu₉ and Cu [34]. Therefore, when an aluminum-copper joint is subjected to a shear force, there is a large stress concentration between Al/Al₂Cu and Al₂Cu/AlCu, which can easily become the starting point of cracks. Furthermore, Al₂Cu has a poor shear resistance and greater mismatch with the adjacent interface, rendering it prone to crack propagation. According to previous experimental results, the Cu content of the Al₂Cu phase is low, and it mainly forms at the transition boundary of the Al-rich zone/Cu-rich zone at the Al-Cu interface. Therefore, the distribution of the Al₂Cu phase is closely related to the invasion depth, d_{Al} . Fig. 14 shows the statistical relationship between the depth of aluminum invasion and shear strength. The results showed that the larger the d_{Al} , the higher the shear strength, with a strong linear correlation (approximately 95%). The shear strength of each group satisfied a linear relationship. That is, the invasion of aluminum into copper increases the degree of anchorage and improves the joint strength. Combining the results of Figs. 9 and 10, as the depth of Al invasion increases, the risk of cracks increases. In the C1280-R sample, when the IMCs on both sides of the interface were only the Al-Cu eutectic and Al₂Cu, the stress concentration was small, and almost only Al₄Cu₉ appeared in the Cu-rich zone. At this time, the joint was insensitive to cracks. Therefore, a core power greater than 1 kW and slightly less than the ring power is beneficial for obtaining a larger penetration depth and d_{Al} and thereby ensure the strength of the joint.

Figs. 15–17 show the shear fracture of the joints with different aluminum invasion depths (d_{Al}). The joints exhibited ductile-brittle mixed fractures, and the fractures of different d_{Al} showed different characteristics. Fig. 15 shows the fracture of C-R0 with a smaller d_{Al} , which is mainly a brittle fracture, and Fig. 15c,e present an obviously smooth plane with cleavage steps and tearing edges at the edge of the plane. There were only a few small dimples in the fracture, and these were mainly distributed at the edge of the Al-Cu interface (Fig. 15b,c). The fracture EDS results showed that a large number of cleavage steps and tearing edges were mainly distributed in Al₂Cu and AlCu phases, and dimples appeared in the Al-Cu eutectic with a high Al content. This indicates that brittle and hard Al₂Cu and AlCu phases easily become areas of crack propagation. During the propagation process, the cracks passed through a part of the Al-Cu eutectic zone to form microdimples.

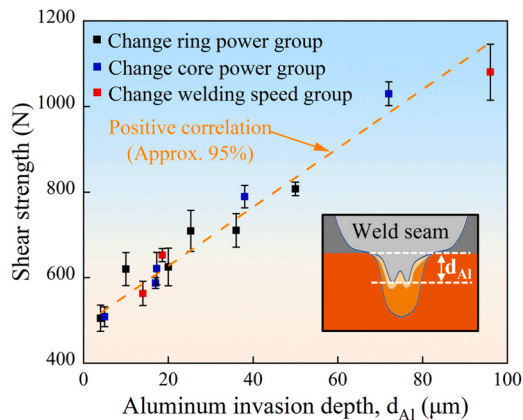


Fig. 14. The relationship between invasion depth d_{Al} and shear strength. The error bar represents the standard deviation of the mean value. The schematic diagram inserted in the bottom right corner is used to describe the meaning of d_{Al} .

Compared with the C-R0 sample, the C1280-R sample exhibited a significantly higher d_{Al} value, and the fracture was a ductile-brittle mixed fracture. The number of dimples increased significantly, and they were mainly distributed in the Al-Cu eutectic and a small amount of the Cu matrix (Fig. 16c,e). Combined with the interface characteristics shown in Fig. 9c,e, the invasion of aluminum into the Cu-rich zone increases the anchoring effect, which results in more ductile zones in during crack propagation and increases the shear resistance of the joint. In addition, Fig. 16a showed that a crack extended from the welding interface to the substrate on the Al side, indicating that the base metal underwent a certain deformation during the fracture process of the joint, and the bonding strength at the interface was close to the strength of the Al substrate.

The fracture of the CR-v200 sample, which has the largest d_{Al} value, is shown in Fig. 17. The number of dimples observed in the fracture of the copper side further increased, and they were mainly distributed in the Al-Cu eutectic. Fig. 10c,e show that the Al₂Cu dendrites formed in the aluminum invasion zone of the Al-Cu interface are larger in size and have larger contact area with the Al-Cu eutectic. Therefore, when cracks propagate along Al₂Cu, more Al-Cu eutectic cracks form, thereby forming more dimples. In addition, more Al₄Cu₉ was detected on both sides of the aluminum-copper fracture, indicating that the depth of crack propagation increased further and that the fracture resistance improved.

According to the shear fracture results, the fracture started at the Al-Cu interface and extended along the IMCs to the middle of the weld until failure occurred. The fracture surface characteristics were determined by the type and distribution of IMCs. To reveal this feature, Fig. 18 shows the crack propagation diagram at the interface when the Al-Cu joint is subjected to a shear force. Fig. 18a shows the formation of an effective anchoring joint at the interface by the metallurgical reaction between Al and Cu, and d_{Al} represents the depth of Al invasion into the Cu-rich zone. There were two evident stress concentrations on both sides of the interface. Fig. 18b shows an enlarged image of the stress concentration area. Under the action of the shear force, the cracks generated by the stress concentration expanded along the path indicated by the blue arrow. Because the Al₂Cu phase was dendritic in the joint, the dendrites intertwined with the Al-Cu eutectic and AlCu phases. Therefore, when the fracture path expanded along Al₂Cu, some cracks extended to the Al-Cu eutectic or even the Al matrix, such that there were obvious dimples in the fracture. This improved the shear strength of the joint. Therefore, ductile fracture caused by crack propagation along the Al₂Cu phase is closely related to the shear strength of the joint. Fig. 18c,d describe the fracture paths of the joints with different aluminum invasion depths. There were two main stages of crack propagation. In the first stage, the initial cracks on both sides of the interface internally extended along the Al₂Cu phase. At this stage, the distribution and morphology of Al₂Cu affected the crack propagation paths. When the d_{Al} value was large, its distribution was closer to the weld boundary and root. Furthermore, the crack propagation path was longer, and the longer fracture path extended to more ductile fracture zones. When the crack extended to the intermediate AlCu and Al₄Cu₉ phases, the second stage of expansion began. At this stage, the bonding strength of the entire joint was reduced significantly, and the cracks rapidly connected and broke. Therefore, the depth of aluminum invasion in Fig. 18c was larger, resulting in a longer fracture path and a larger ductile fracture zone. This increased the failure resistance and shear strength.

5. Conclusion

In this study, the microstructures and properties of 1060 pure aluminum and T2 pure copper adjustable ring-mode (ARM) laser lap joints were investigated. The shear strengths and electrical conductivities of the joints were compared at different power ratios and welding speeds. The results indicated that this process can achieve an effective connection and obtain welded joints without obvious defects. Al and Cu exhibited different flow behaviors under different adjustable ring-mode

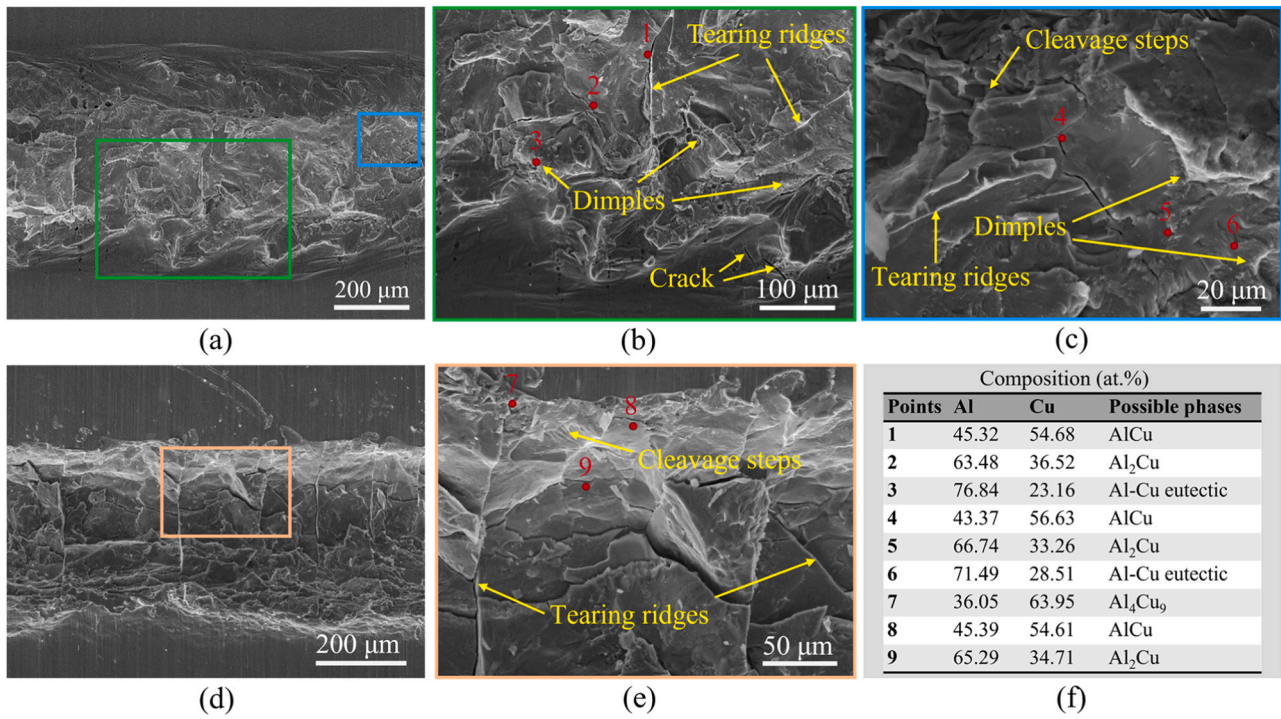


Fig. 15. The fracture morphology of C-R0. (a-c) The fracture of Al side. (d-e) The fracture of Cu side. (f) Elemental distribution of the fracture.

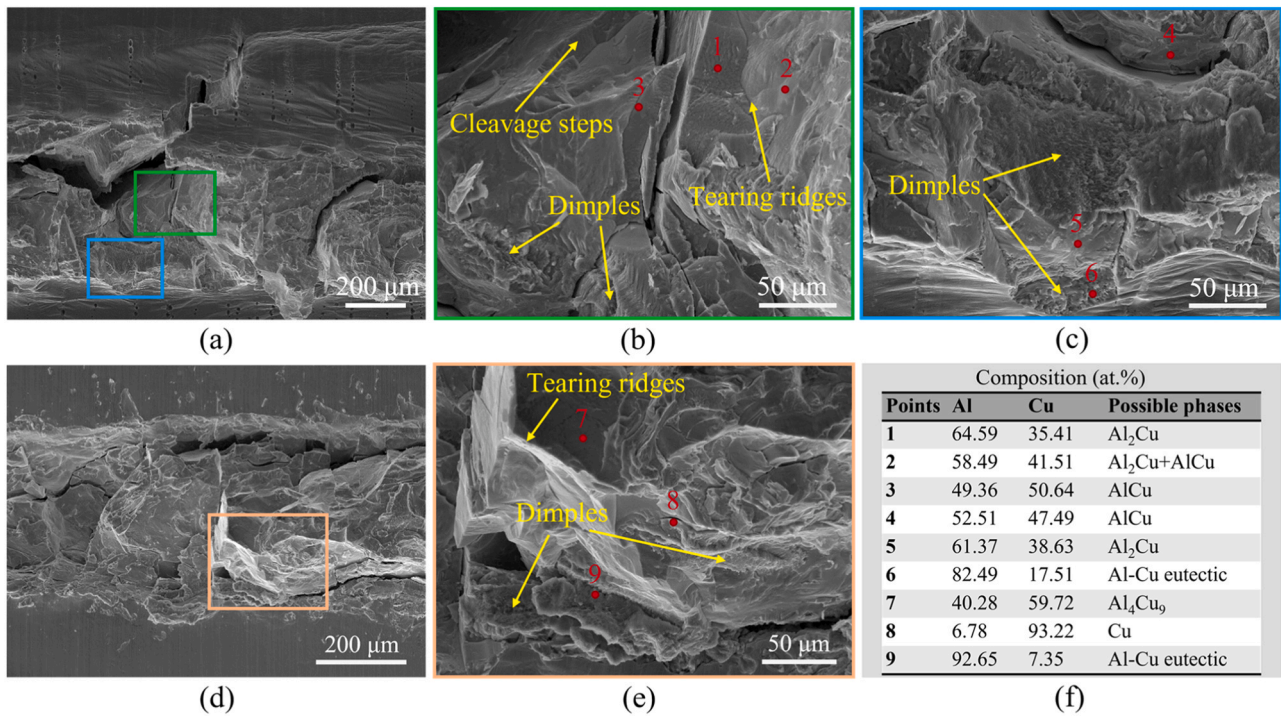


Fig. 16. The fracture morphology of C1280-R. (a-c) The fracture of Al side. (d-e) The fracture of Cu side. (f) Elemental distribution of the fracture.

(ARM) laser energy distributions, leading to different weld formation mechanisms, microstructure, and connection strength. The main conclusions of this study are as follows:

- (1) The core and ring lasers have different effects on the weld. The core power has a significant impact on the weld penetration and width, whereas the ring power has a certain effect on the

formation of welds. As the power ratio (r) decreased, the surface roughness and spatter of the weld decreased.

- (2) Under the force of the molten pool, copper at the bottom flowed upward and eventually distributed within the Al-Cu mixing zone in the form of vortices. The ring laser caused the upper aluminum to flow deeper along the eddy currents from both sides of the molten pool, ultimately forming an aluminum intrusion zone. By adjusting the adjustable ring-mode (ARM) laser, a larger joint

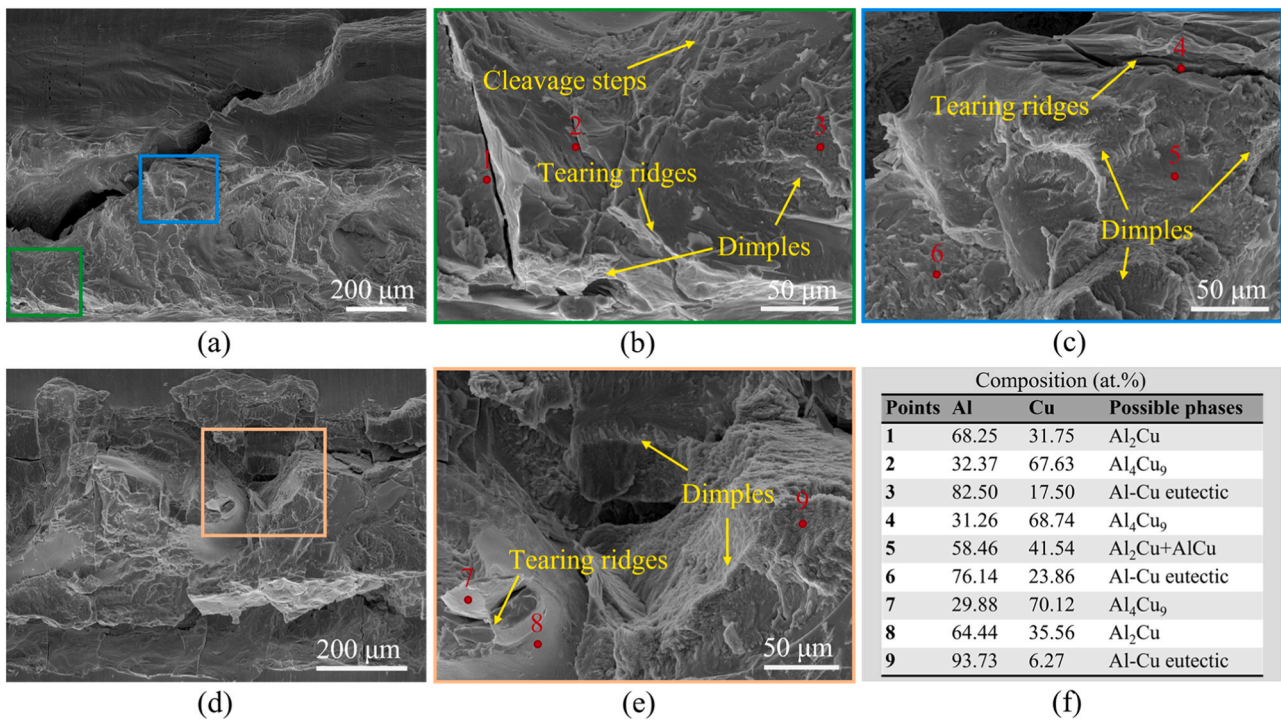


Fig. 17. The fracture morphology of CR-v200. (a-c) The fracture of Al side. (d-e) The fracture of Cu side. (f) Elemental distribution of the fracture.

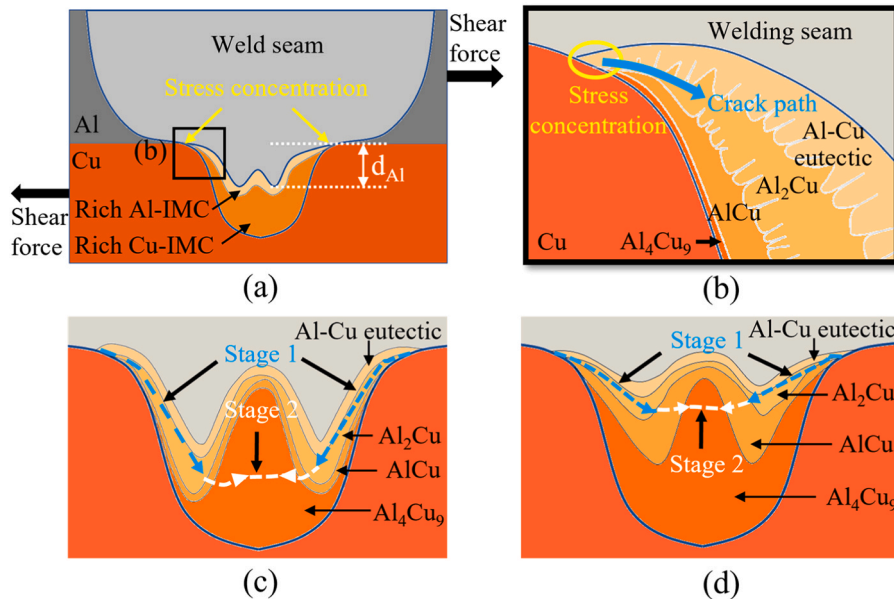


Fig. 18. The schematic diagram of crack propagation at the interface. (a) Interface stress concentration. (b) Local enlarged image. (c-d) Fracture paths under different Al invasion depths.

depth was achieved, and the mixing of Al and Cu was controlled in the Al-Cu mixing zone at the upper part of the weld, which limited element mixing in the Cu-rich zone.

- (3) The Al-Cu interface mainly formed three intermetallic compounds (IMCs): Al₂Cu, AlCu, and Al₄Cu₉. Al₂Cu was dendritic and mainly distributed near the aluminum side, whereas Al₄Cu₉ was concentrated in the copper melting zone. These two types of IMCs preferentially formed, and an AlCu phase was also formed.
- (4) The shear strength of the joint changed from 508 N to 1080 N, and the strength linearly and positively correlated with the depth of aluminum invasion (d_{Al}). The shear fracture exhibited obvious

ductile-brittle mixed characteristics. The cracks mainly propagated along Al₂Cu, and during the propagation stage, the Al-Cu eutectic and partial matrix cracking formed dimples. The resistance value of the joint decreases from 66.7 μΩ to 53 μΩ with an increase in the interface connection width.

CRediT authorship contribution statement

Hao Dong: Methodology, Investigation, Formal analysis. **Yan Cai:** Writing – review & editing, Supervision, Resources, Project administration. **Bin Zhang:** Visualization, Data curation. **Shuangyang Zou:**

Investigation, Data curation. **Wucheng Li**: Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Weidong Mu**: Supervision, Resources.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] Tokita, S., Liu, C.Y., Sato, Y.S., 2023. Effects of material combination for ultrasonic spot welding on microstructure and strength of Al/Al, Al/Cu, and Al/Ni dissimilar joints. *J. Mater. Res. Technol.* 26, 7619–7633.
- [2] Lee, K., Rinker, T.J., Pour, M.M., Cai, W., Huang, W.K., Tan, W.D., Bracey, J., Li, J. J., 2023. A study on cracks and IMCs in laser welding of Al and Cu. *Manuf. Lett.* 35, 221–231.
- [3] Ma, D., Jiang, P., Shu, L., Geng, S., 2022. Real-time porosity monitoring during laser welding of aluminum alloys based on keyhole 3D morphology characteristics. *J. Manuf. Syst.* 65, 70–87.
- [4] You, J.Q., Zhao, Y.Q., Dong, C.L., Miao, S., Liu, Z., Liu, L., Su, Y.H., 2022. Microstructural evolution and mechanical properties of the Al-Cu dissimilar joint enhanced by stationary-dynamic shoulder friction stir welding. *J. Mater. Process. Technol.* 300, 117402.
- [5] Xue, P., Ni, D.R., Wang, D., Xiao, B.L., Ma, Z.Y., 2011. Effect of friction stir welding parameters on the microstructure and mechanical properties of the dissimilar Al-Cu joints. *Mater. Sci. Eng. A* 528 (13), 4683–4689.
- [6] Mathivanan, K., Plapper, P., 2019. Laser welding of dissimilar copper and aluminum sheets by shaping the laser pulses. *Procedia Manuf.* 36, 154–162.
- [7] Zhou, L., Li, Z.Y., Song, X.G., Tan, C.W., He, Z.Z., Huang, Y.X., Feng, J.C., 2017. Influence of laser offset on laser welding-brazing of Al/brass dissimilar alloys. *J. Alloy. Compd.* 717, 78–92.
- [8] Schmalen, P., Plapper, P., Peral, I., Titov, I., Vallcorba, O., Rius, J., 2018. Composition and phases in laser welded Al-Cu joints by synchrotron x-ray microdiffraction. *Procedia CIRP* 74, 27–32.
- [9] Mao, Y., Qin, D., Xiao, X., Wang, X., Fu, L., 2023. Achievement of high-strength Al/Cu dissimilar joint during submerged friction stir welding and its regulation mechanism of intermetallic compounds layer. *Mater. Sci. Eng. A* 865, 144164.
- [10] Dang, Z., Qin, G., Li, T., 2023. Microstructure evolution and tensile strength of Al/Cu inertia friction welded joint. *J. Mater. Res. Technol.* 27, 4023–4031.
- [11] Wang, X., Li, X., Wang, C., 2018. Influence of diffusion brazing parameters on microstructure and properties of Cu/Al joints. *J. Manuf. Process.* 35, 343–350.
- [12] Ma, Q., Song, C., Zhou, J., Zhang, L., Ji, H., 2021. Dynamic Weld evolution during ultrasonic welding of Cu-Al joints. *Mater. Sci. Eng. A* 823, 141724.
- [13] Veeresh, P., Wang, H., Wang, H., Oswald, J., 2023. A computational and experimental study in resistance spot welding of thin-gauge and highly conductive dissimilar sheets for battery tab interconnects. *J. Mater. Process. Technol.* 322, 118192.
- [14] Tanmay, Panda, S.S., 2021. Characterisation of Cu-Al alloy lap joint using TIG Welding. *Cirp. J. Manuf. Sci. Technol.* 35, 454–459.
- [15] Li, F.S., Wei, Z.C., Li, X.F., Kang, H.D., Li, Z., Xu, R.Z., 2020. Investigation on the coupling interaction in electron beam welded Al-Cu bimetallic sheet. *P. I. Mech. Eng. L-J. Mat.* 235 (3), 640–652.
- [16] Hollatz, S., Hummel, M., Olowinsky, A., Gillner, A., Beckmann, F., Moosmann, J., 2022. Pore formation and melt pool analysis of laser welded Al-Cu joints using synchrotron radiation. *J. Mater. Process. Technol.* 309, 117738.
- [17] Yan, S., Shi, Y., 2020. Influence of Ni interlayer on microstructure and mechanical properties of laser welded joint of Al/Cu bimetal. *J. Manuf. Process.* 59, 343–354.
- [18] Huang, W., Cai, W., Rinker, T.J., Bracey, J., Tan, W., 2023. Effects of laser oscillation on metal mixing, microstructure, and mechanical property of Aluminum-Copper welds. *Int. J. Mach. Tools Manuf.* 188, 104020.
- [19] Liu, F., Xu, B., Song, K., Tan, C., Zhao, H., Wang, G., Song, X., 2022. Improvement of penetration ability of heat source for 316 stainless steel welds produced by alternating magnetic field assisted laser-MIG hybrid welding. *J. Mater. Process. Technol.* 299, 117329.
- [20] Farhang, B., Tanrikulu, A.A., Ganesh-Ram, A., Jain, A., Amerinatanzi, A., 2023. Electromagnetic field-assisted laser welding of NiTi to stainless steel: Towards a lightweight, high-strength joint with preserved properties. *J. Mater. Process. Technol.* 314, 117888.
- [21] Ma, B., Gao, X., Huang, Y., Zhang, Y., Huang, Y., 2024. Effect of different pulse shapes on the laser welding of aluminum and copper. *Opt. Laser Technol.* 171, 110312.
- [22] Maina, M.R., Okamoto, Y., Okada, A., Närhi, M., Kangastupa, J., Vihinen, J., 2018. High surface quality welding of aluminum using adjustable ring-mode fiber laser. *J. Mater. Process. Technol.* 258, 180–188.
- [23] Cha, J.H., Choi, H.W., 2022. Characterization of dissimilar aluminum-copper material joining by controlled dual laser beam. *Int. J. Adv. Manuf. Technol.* 119 (3), 1909–1920.
- [24] Wan, L., Tang, Z., Yang, H., Sun, H., Wei, Q., Wu, Y., Wang, H., 2024. Achieving high robust laser conduction welding and enhanced joint conductivity in pure copper foil stacks using a flat-top blue laser. *Opt. Laser Technol.* 174, 110701.
- [25] Jabbar, S., Baghbani Barenji, A., Franciosa, P., Kotadia, H.R., Ceglarek, D., 2023. Effects of the adjustable ring-mode laser on intermetallic formation and mechanical properties of steel to aluminum laser welded lap joints. *Mater. Des.* 227, 111774.
- [26] Shin, J., Kang, S., Hyun, K., Ji, H., 2022. In-depth characterization of an aluminum alloy welded by a dual-mode fiber laser. *J. Mater. Res. Technol.* 18, 2910–2921.
- [27] Huang, W., Wang, H., Rinker, T., Tan, W., 2020. Investigation of metal mixing in laser keyhole welding of dissimilar metals. *Mater. Des.* 195, 109056.
- [28] Muhammad, N.A., Wu, C.S., Tian, W., 2019. Effect of ultrasonic vibration on the intermetallic compound layer formation in Al/Cu friction stir weld joints. *J. Alloy. Compd.* 785, 512–522.
- [29] Haidara, F., Mangelinck, D., Duployer, B., Record, M., 2012. Phase formation of Al10Cu10Fe in thin films. *Mater. Chem. Phys.* 133 (2), 977–980.
- [30] Song, G., Li, T.T., Chen, L., 2018. The mechanical properties and interface bonding mechanism of immiscible Mg/steel by laser-tungsten inert gas welding with filler wire. *Mater. Sci. Eng. A* 736, 306–315.
- [31] Peng, P., Wang, K., Wang, W., Yang, T., Liu, Q., Zhang, T., Liu, J., 2021. Intermetallic compounds: Formation mechanism and effects on the mechanical properties of friction stir lap welded dissimilar joints of magnesium and aluminum alloys. *Mater. Sci. Eng. A* 802, 140554.
- [32] He, W., Ma, H., Shi, Z.H., Wang, L.M., Ke, D.F., Tang, C., Xiao, W., Xue, B.Y., Wang, J.W., 2022. Theoretical study on segregation behavior of grain boundaries and precipitated phases in Al-Cu alloys. *Mater. Sci. Technol.* 30 (02), 68–74.
- [33] Yu, H., Zhang, L., Cai, F., Zhong, S., Ma, J., Zhang, Y., Osaka, A., 2020. Interface microstructure and growth mechanism of brazing Cu/Al joint with BAl88Si filler metal. *Vacuum* 181, 109641.
- [34] Rabkin, D.M., Ryabov, V.R., Lozovskaya, A.V., Dovzhenko, V.A., 1970. Preparation and properties of copper-aluminum intermetallic compounds. *Powder Metall. Met. Ceram.* 9 (8), 695–700.